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vSmart insights on vManage

Created by Raghu Chereddy, last modified by Ramana Polamarasetti on May 27, 2025

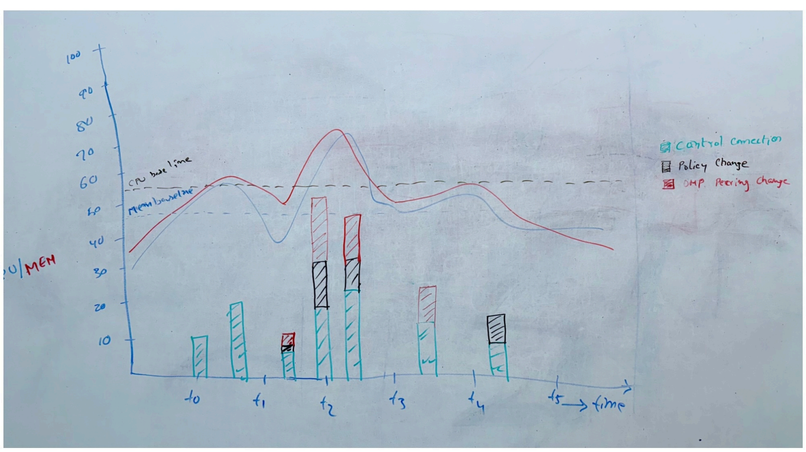
Problem statement:

When OMP peering is lost, there is no easy way to find out the reason. The OMP peering can be lost due to several reasons:

- High CPU/Memory
- Loss of vDaemon control connection
- Expiration of OMP hold timer

Users need to analyze several log files to understand which of the above caused loss of OMP peering. While engineers can analyze, it would be a challenge for customers, TAC and Escalation teams.

The intent of this feature is to collect required data and present it to the users, to understand the cause of OMP peering failures.



Components:

There are three components that need to be implemented.

- Data collection on vSmart
- Data transfer to vManage
- Data presentation

Each of these components are described below.

vSmart development branch:

feature/vsmart_insights

Data collection on vSmart:

vSmart collects the data periodically, every 3 minutes (TBD, based on test data). The data is written to a PSV and the format is described below.

This data will be captured from OMP data structures and sysmgr (CPU/Memory).

Please refer to the following link for vSmart global and the events data.

https://cisco-my.sharepoint.com/:x:/r/personal/rpolamar_cisco_com/_layouts/15/Doc.aspx?sourcedoc=%7B400AF266-F61C-44D2-9819-47996AC62E30%7D&file=vsmart_vmanage_insights.xlsx&wdLOR=cCBCCBBEA-8B50-FF4D-9AC1-AF0A0571627F&fromShare=true&action=default&mobileredirect=true

vSmart writes global data into PSV files. The PSV data format is as below:

```
Time Stamp | Tenant ID | Site ID | System IP | Num of CC Flaps | Num of OMP Peer Flaps | Routes Sent | Routes Received | CPU | Memory
```

vSmart sends the events data to the vManage, whenever an event occurs. In this case, the event would be OMP peering state change. The following fields will be added to the existing events data:

- Routes sent (to this peer)
- Routes received (from this peer)
- OMP uptime
- CC uptime

The above information will be sent part of "viptela_omp_omp_peer_state_change" type.

Data transfer to vManage:

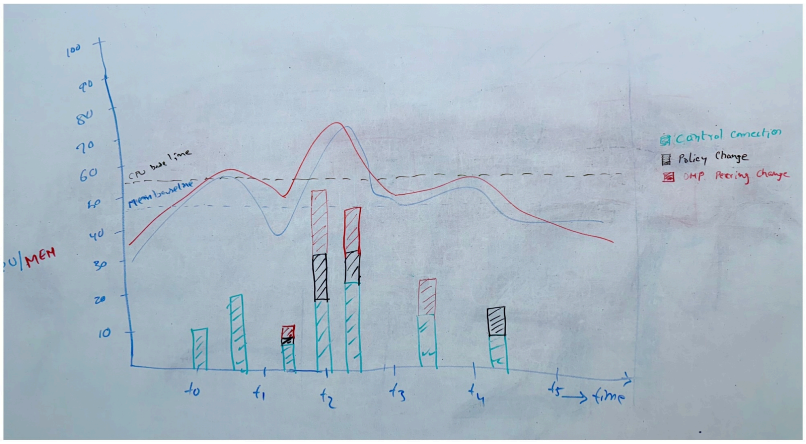
The PSV files are pushed by the vSmart periodically, every 10 minutes.

Data presentation:

4/8/2025

1. Push model will be implemented on the vSmart. Along with vSmart insights data, all existing data will be transitioned to push model.
2. All graphs discussed earlier will be consolidated into one time series graph.
3. The graph will include Memory, CPU usage.
4. The periodic statistical data points will be plotted on the same time series graph.
5. Peer events will be shown on the chart as a bar graph. Both periodic statistical data and the events bar chart will be shown on the same graph.
6. The peer events, will be displayed as a table if user selects to zoom in between two time stamps.
7. Users will have the ability to filter data based on peer IP, Site ID, and event type. When a filter is applied, the chart and table will have data corresponding to the filters only.

The following is graph showing the periodic data and the events.



vManage support / implementation

vSmart will use the following REST API to upload the PSV files to vManage. The DeviceDataCollector (DDC) uses port 8129. Along with the file to be uploaded, vSmart must also send the 'stat type', 'file name' and 'tenant org name' in the HTTP header as shown in the examples below:

curl http://169.254.10.1:8129/datacollection/v1/ -H "X-FileName:vsmart-apple-1744594011.psv.gz" -H "X-FileClass:vsmart" -H "X-TenantOrgName:DCA AUT VMANAGE-Apple Inc" -F "uploadFile=@vsmart-apple-1744594011.psv.gz"

Here is a sample vsmart-apple-1744594011.psv.gz:

```
entry_time|tenant_org_name|site_id|system_ip|host_name|num_cc_flaps|num_omp_peer_flaps|routes_sent|routes_received|cpu|memory
1627849320|DCA AUT VMANAGE-Apple Inc|129|172.16.253.129|vm129|15|25|35|45|65|85
```

curl http://169.254.10.1:8129/datacollection/v1/ -H "X-FileName:vsmart-mango-1744594011.psv.gz" -H "X-FileClass:vsmart" -H "X-TenantOrgName:DCA AUT VMANAGE-Mango Inc" -F "uploadFile=@vsmart-mango-1744594011.psv.gz"

Here is a sample vsmart-mango-1744594011.psv.gz:

```
entry_time|tenant_org_name|site_id|system_ip|host_name|num_cc_flaps|num_omp_peer_flaps|routes_sent|routes_received|cpu|memory
1627849339|DCA AUT VMANAGE-Mango Inc|129|172.16.253.129|vm129|10|20|30|40|66|88
```

No labels